

Bhavya Vashisht

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Education

Thapar Institute Of Engineering and Technology Patiala
Bachelor of Engineering in Electronics and Communication Engineering 2022 – 2026
CGPA: 8.75

Shalom Hills International School Gurugram
CLASS XII (PCM with CS) CBSE 95.8% 2020 – 2021

Experience

Yum! Brands – Digital Food Platform Gurugram
Machine Learning Engineer Intern January, 2026 – Present

- Built production recurring agent scheduling on Kubernetes CronJobs — REST schedule API with 7 recurrence patterns, CronJob lifecycle manager (`concurrencyPolicy: Forbid`), and a worker that orchestrates agent/team invocation and writes run records to Snowflake.
- Designed Redis-backed one-time scheduling with at-least-once delivery, an idempotency guard across horizontally scaled worker replicas, and a single-replica Snowflake→Redis sync coordinator so DB read load stays constant as workers scale — periodic reconcile for reliable production execution after Redis failover or wipe.
- Shipped async LLM cost tracking: non-blocking emit after each chat turn, Redis ZSET job queue, background worker with retry/backoff, and Snowflake model-pricing cache — chat responses never wait on cost calculation.
- Built production K8s deployment infrastructure for agents and multi-agent teams — unified REST deploy API (create/save/redeploy/undeploy), Helm lifecycle with Snowflake status tracking, init-container GitLab preload for runtime actions, and deployment-scoped ingress + asset URLs for isolated pod routing.
- Integrated Model Context Protocol (stdio + Streamable HTTP); fixed a production race condition in shared MCP config via request-scoped deep-copy across 3 execution paths.
- Optimized a MongoDB conversation-list API via metadata-only endpoint + compound index — payload 1.6 MB → 7 KB (99.5%), latency 6–15s → sub-500ms.

Skills

Languages: Python, Go, C, SQL

Frameworks & Libraries: FastAPI, PydanticAI, MCP, JWT

Databases & Caching: MongoDB, PostgreSQL, Redis

Tools: Git, Docker, Kubernetes, Helm, GitLab CI/CD, Linux, SQLC, Goose

Coursework: Data Structures & Algorithms, OS, DBMS, OOP, AI, Computer Architecture

Certifications: HTTP Clients in Go, Learn Linux, Learn OOP in Python, Learn Go

Projects

Deputy | Python, Ollama, MCP, sqlite-vec, FastAPI [GitHub](#)

- Built a private on-device AI agent: a bounded ReAct loop with constrained decoding that lifted end-to-end task success 29%→88% on `qwen2.5:3b` (schema-valid steps 71%→100%), with tools exposed over MCP.
- Added on-device RAG (sqlite-vec + Ollama embeddings) and a trust surface — approval gates on writes, an append-only audit log, and local-first routing with opt-in cloud; local FastAPI UI, reliability eval harness, 189 tests, strict mypy.

Chirpy | Go, PostgreSQL, JWT, SQLC, Goose [GitHub](#)

- Built a REST API in Go for a micro-blogging platform with JWT auth, bcrypt hashing, and middleware for logging, validation, and error recovery.
- Designed a PostgreSQL schema with SQLC + Goose (5 migrations, 20+ queries); added Chirpy Red premium upgrades via webhooks.

JobOps Pipeline | Python, Google Sheets API, asyncio/httpx, LLM APIs [GitHub](#)

- Built a daily job-ops pipeline (~8.6k LOC): `morning.py` orchestrates Sheet sync → parallel fetch across 15+ job boards and ATS APIs → fresher/geo/salary filters → tiered scoring → bidirectional Google Sheets CRM.
- Implemented dedup by URL plus company/role/location collapse; optional LLM JD enrichment (Gemini/OpenAI) extracts stack/YOE and auto-picks resume variant; cached in `.jd_cache.json`.
- Added outreach step: Hunter.io contact lookup, per-role cold-mail drafts, and company referral DMs — review-before-send workflow (never auto-sends).